



# Building an E-commerce Presence: Web Sites, Mobile Sites, and Apps

# E-commerce Presence

**E-commerce presence** refers to the **online visibility and availability** of a business or organization that sells products or services over the internet. It includes all the **digital platforms and channels** through which a business interacts with customers, showcases its offerings, and conducts transactions.

## Key Components of an E-commerce Presence:

### 1. Website (Main Hub)

1. A professional e-commerce website is the core of your online presence.
2. Example: An online clothing store with product pages, a shopping cart, and checkout functionality.

### 2. Mobile Presence

1. Mobile-optimized websites or dedicated mobile apps.
2. Example: A food delivery app that allows users to place orders on their smartphones.

# E-commerce Presence

## 3. Social Media

- 3. Platforms like Facebook, Instagram, YouTube, or TikTok used for marketing, customer engagement, and even direct selling.
- 4. Example: A cosmetics brand selling products through Instagram Shops and promoting via influencers.

## 4. Email Marketing

- 3. Direct communication with customers via newsletters and promotional emails.
- 4. Example: Sending weekly product recommendations or special discount codes to subscribers.

## 5. Search Engine Presence

- 3. Visibility through search engines like Google, using SEO and paid ads (e.g., Google Ads).
- 4. Example: An electronics store appearing on the first page of Google when someone searches "buy laptop online."

## 6. Online Marketplaces

- 3. Selling through third-party platforms like Amazon, eBay, or Etsy.
- 4. Example: A small business selling handmade crafts on Etsy.



# Why E-commerce Presence is Important?

- ❖ **Reach a wider audience** beyond physical boundaries.
- ❖ **Improve brand visibility** and trust.
- ❖ **Increase sales** by making it easy for customers to find and buy products.
- ❖ **Enhance customer engagement** through multiple digital touchpoints.

# Reach a Wider Audience Beyond Physical Boundaries

In a traditional brick-and-mortar setup, your business is limited by geography. However, an e-commerce presence removes these barriers and allows you to:

- ❖ **Sell to customers worldwide**, 24/7, regardless of location or time zone.
- ❖ **Tap into new markets** without the cost of setting up physical stores.
- ❖ **Cater to niche segments** that may not exist in your local area but have strong demand elsewhere.

## Example:

A boutique in Rajshahi can now sell handcrafted sarees to customers in Dhaka, Dubai, or even New York through an online store.

# Improve Brand Visibility and Trust

Having a professional and active digital presence builds **brand credibility** and keeps your business top-of-mind for consumers. Here's how:

- ❖ A well-designed website, consistent branding, and active social media pages signal that your business is **legitimate and trustworthy**.
- ❖ Positive online reviews, testimonials, and user-generated content increase **social proof**.
- ❖ Search engine optimization (SEO) ensures that customers **find your brand when they search online**.

## Example:

If someone searches "organic honey in Bangladesh," your brand appears in the top search results due to a strong web presence—instantly improving visibility and trust.



# Increase Sales by Making it Easy for Customers to Find and Buy Products

A strong e-commerce presence simplifies the shopping process and boosts conversion rates:

- ❖ Customers can **search, compare, and purchase** products in just a few clicks.
- ❖ Features like personalized recommendations, shopping carts, and secure payment gateways make it **convenient and efficient**.
- ❖ You can run **targeted promotions** and offer discounts to attract and retain buyers.

## Example:

A user browsing Instagram sees a sponsored ad for your t-shirt brand, clicks the link, views product options on your website, and checks out within minutes—all without ever visiting a store.

# Enhance Customer Engagement Through Multiple Digital Touchpoints

With an e-commerce presence, businesses can connect and interact with customers through various channels, leading to stronger relationships:

- ❖ **Social media** lets you run polls, reply to comments, or share behind-the-scenes content.
- ❖ **Live chat or chatbots** on your website allow real-time support.
- ❖ **Email newsletters** keep customers informed about new arrivals, sales, and personalized offers.
- ❖ **Example:**

A customer adds an item to their cart but doesn't complete the purchase. Later, they receive an automated email reminding them to check out—this small engagement could lead to a sale.



# Imagine Your E-commerce Presence

## Determine the vision of your Business

### What is the idea?

Before building an e-commerce presence, it's essential to have a clear vision. This vision should outline your mission, target audience, market space, strategic analysis, marketing plan, and a development timeline with a preliminary budget. It begins with an idea of what's possible and ends with a practical plan for implementation.

# What is the idea?

## Example:

Imagine a startup planning to launch an e-commerce store that sells eco-friendly clothing. Before building the website, the founders develop a clear vision:

- ❑ **Mission:** To make sustainable fashion affordable and accessible to everyone.
- ❑ **Target Audience:** Environmentally conscious young adults aged 18–35 in urban areas.
- ❑ **Market Space:** The sustainable fashion niche within the broader online apparel industry, focused initially on North America.

# What is the idea?

- ❑ **Strategic Analysis (SWOT):**
  - ❖ **Strengths:** Unique eco-friendly materials, transparent supply chain.
  - ❖ **Weaknesses:** New brand with no established reputation.
  - ❖ **Opportunities:** Growing demand for ethical fashion.
  - ❖ **Threats:** Competition from established brands going green.
- ❑ **Marketing Matrix:** Using Instagram and TikTok for awareness, influencers for trust-building, and targeted ads for conversions.
- ❑ **Development Timeline & Budget:** A 6-month plan including design, product sourcing, marketing, and launch, with an estimated budget of \$50,000.

With this vision in place, the team avoids missteps, stays aligned, and executes their e-commerce presence effectively.



# Where's the \$\$money?

To build a sustainable e-commerce presence, you need to be clear on:

## **Business model(s):**

Portal, e-tailer, content provider, transaction broker, market creator, service provider, community provider

## **Revenue model(s):**

Advertising, subscriptions, transaction fees, sales, and affiliate revenue

## **Why It Matters:**

Clearly defining **where the money comes from** helps:

- ❖ Build a solid business plan
- ❖ Attract investors or partners
- ❖ Set pricing strategies
- ❖ Forecast future revenue and growth

# Who and where is the target audience?

Understanding your audience is key to delivering the right product, message, and experience. Below are the core dimensions for profiling your e-commerce audience:

## Demographics

Demographics provide **basic statistical data** about your potential customers. These are often the first criteria used to segment and define an audience.

### Key Elements:

- ❖ **Age:** Are they teens (13–19), millennials (20–35), adults (36–55), or seniors (55+)?
- ❖ **Gender:** Are your offerings more appealing to men, women, or all genders?
- ❖ **Income:** Determines affordability and purchasing behavior (low, middle, or high-income segments)
- ❖ **Location:** Urban vs. rural; local, national, or global; cultural context may vary

# Characterize the marketplace

To **imagine your e-commerce presence** effectively, one important step is to **characterize the marketplace** where you will operate. This involves analyzing the environment in which your business will function — understanding the market's structure, key players, and dynamics.

## Demographics

This refers to the statistical characteristics of your target market or customer base, which influence buying behavior.

- ❖ **Age:** Are you targeting teens, adults, or seniors?
- ❖ **Gender:** Is your product more appealing to males, females, or all genders?
- ❖ **Income Level:** Are you selling premium or budget-friendly items?
- ❖ **Education Level, Occupation, Lifestyle:** These help shape marketing tone, product design, and platform usability.

**Example:** A platform selling premium smartwatches might target tech-savvy, urban professionals aged 25–45 with mid-to-high income.



# Characterize the marketplace

## Size, Growth, Changes

Understand the **current market size**, whether it is **growing, stagnant, or declining**, and what trends are influencing change.

- ❖ **Size:** What is the estimated number of potential customers or total revenue of the market?
- ❖ **Growth Rate:** Is the market expanding year over year?
- ❖ **Trends:** Are there technological shifts, changing customer expectations, or regulatory changes?

**Example:** The global online grocery market is growing due to increased smartphone use and convenience culture.

## Structure

Defines how the market is organized and who the dominant players are.

- ❖ **Is the market fragmented or monopolized?**
- ❖ **Is there room for new entrants?**
- ❖ **How easy is it for a customer to switch between competitors?**

**Example:** The ride-sharing market is dominated by Uber and Lyft, but smaller regional apps can still find a niche.

# Characterize the marketplace

## Competitors

Identify direct and indirect competitors and assess their strengths and weaknesses.

- **What are they offering?**
- **What's their market share?**
- **How do they price and promote their products?**
- **What is their unique value proposition?**

**Example:** For an online bookstore, Amazon is a direct competitor. Indirect competitors could be audiobook platforms like Audible.

# Characterize the marketplace

## Suppliers

Suppliers provide the goods or services you will sell or use to operate your platform.

- **Are suppliers local or international?**
- **Do they offer reliable delivery and quality?**
- **How dependent is your business on them?**
- **Can you switch suppliers easily?**

**Example:** A dropshipping e-commerce site might rely on Chinese manufacturers through platforms like AliExpress.



# Characterize the marketplace

## Substitute Products

These are alternative products or services that can meet the same customer needs.

- **Are customers already solving this problem differently?**
- **What would convince them to switch to your offering?**

**Example:** Instead of buying gym equipment online, customers may choose digital fitness subscriptions like Peloton.

# Characterize the marketplace

## Where is the Content Coming From?

The **source and nature of your website content** plays a big role in user engagement and SEO.

- ❖ **User-generated:** Reviews, photos, comments (e.g., Amazon, Yelp).
- ❖ **Business-generated:** Product descriptions, blog posts, videos.
- ❖ **Automated feeds:** Affiliate or dropshipping databases.

**Example:** A fashion store may use both in-house editorial content and customer reviews to build trust.

# Characterize the marketplace

## Static or Dynamic Web Pages?

The type of content delivery affects website performance, flexibility, and user experience.

- ❖ **Static pages:** Fixed content (about pages, terms of service). Fast and simple.
- ❖ **Dynamic pages:** Content is generated in real-time based on user interaction or database entries (product pages, user dashboards, personalized recommendations).

**Example:** A product page that changes price based on location or shows “items left in stock” is dynamic.

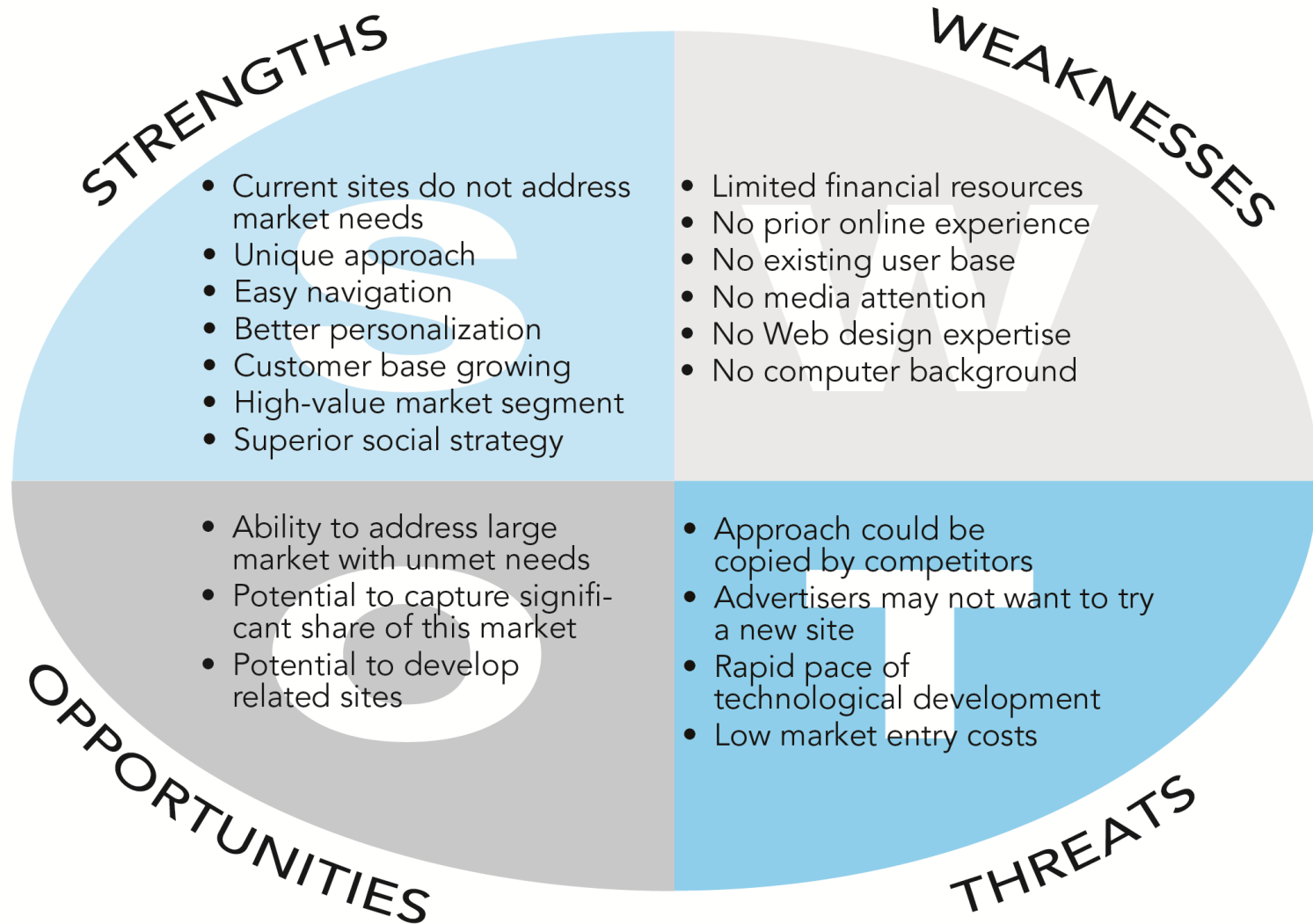


# Know yourself—SWOT analysis

"Know yourself — SWOT analysis" in the context of **e-commerce presence** means evaluating your online business's internal strengths and weaknesses, as well as the external opportunities and threats it faces. This helps you understand your current position, make strategic decisions, and improve your competitiveness in the digital marketplace.

- ❖ **Strength**
- ❖ **Weakness**
- ❖ **Opportunities**
- ❖ **Threats**

# SWOT Analysis



# Know yourself—SWOT analysis

## Strengths (Internal, Positive Factors)

These are the internal capabilities and advantages that give your e-commerce business a competitive edge.

### Examples:

- ❖ **Strong Brand Recognition:** A popular fashion e-commerce site like **Zara.com** benefits from high brand trust and repeat customers.
- ❖ **Efficient Supply Chain:** **Amazon** has an advanced logistics and fulfillment system, enabling fast delivery.
- ❖ **User-Friendly Website or App:** A clean, fast-loading interface with smooth navigation increases user satisfaction (e.g., **Shopify-powered stores**).
- ❖ **Diverse Product Range:** Offering a wide range of products increases customer retention (e.g., **AliExpress**).



# Know yourself—SWOT analysis

## Weaknesses (Internal, Negative Factors)

These are limitations within the business that hinder growth or reduce performance.

### Examples:

- ❖ **Poor Website UX/UI:** A cluttered or slow website can lead to high bounce rates.
- ❖ **Limited Payment Options:** If your e-commerce store doesn't support mobile payments or local currencies, it may lose customers.
- ❖ **Low Inventory Management:** Inaccurate stock data can cause frequent "out-of-stock" messages, frustrating customers.
- ❖ **Lack of Digital Marketing Expertise:** Without effective SEO, SEM, or social media strategies, your business may struggle to attract traffic.

# Know yourself—SWOT analysis

## Opportunities (External, Positive Factors)

These are favorable external conditions that your business can exploit to gain advantage.

### Examples:

- ❖ **Growing Mobile Commerce:** More users shopping via smartphones—optimize for mobile to boost conversions.
- ❖ **New Market Segments:** Selling eco-friendly products in regions with rising environmental awareness.
- ❖ **Government E-commerce Incentives:** For instance, digital infrastructure support in developing countries.
- ❖ **Global Reach:** With the internet, you can easily expand beyond local markets (e.g., using platforms like **Etsy** or **Amazon Global**).

# Know yourself—SWOT analysis

## Threats (External, Negative Factors)

These are external elements that could cause problems for your e-commerce business.

### Examples:

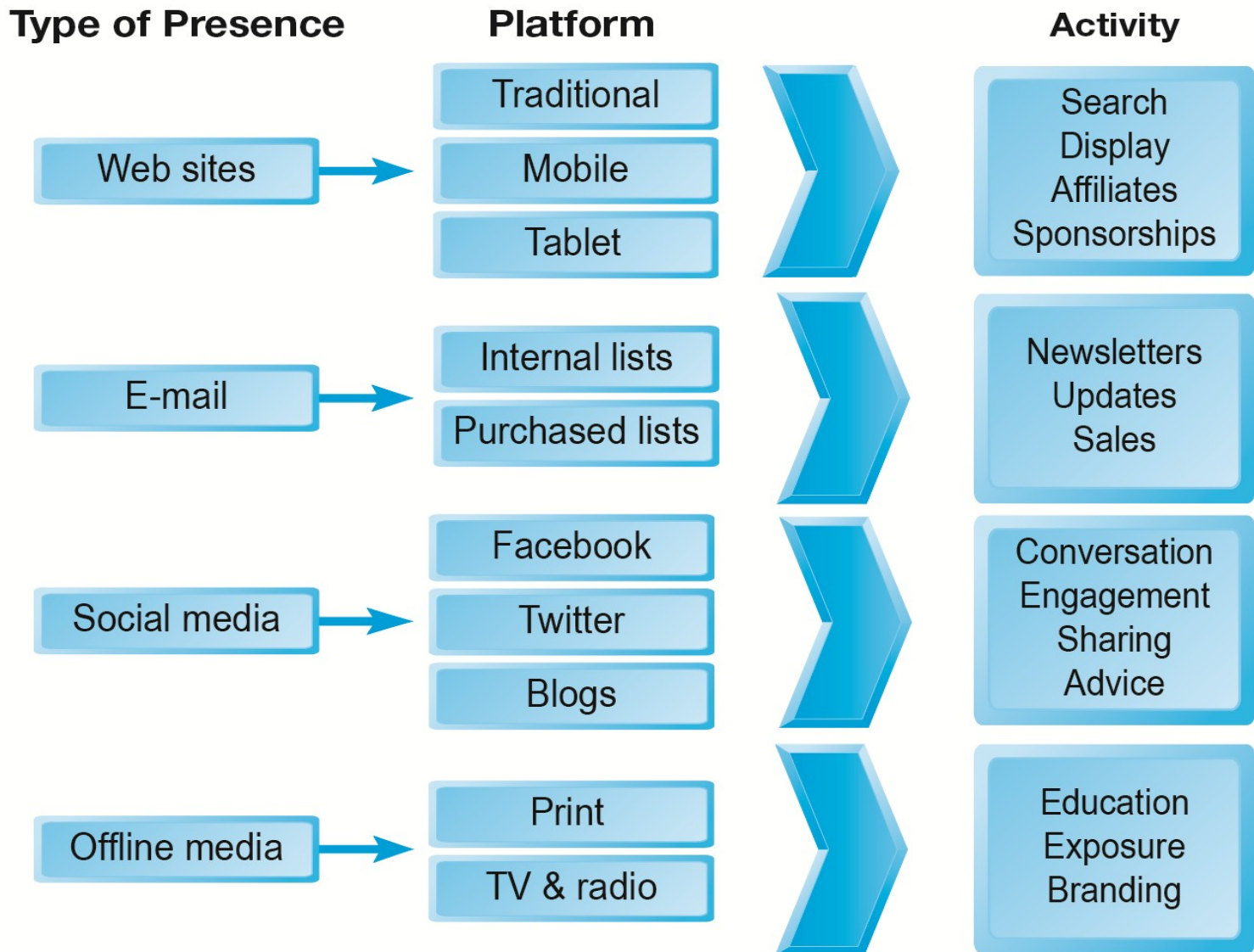
- ❖ **Intense Competition:** Big players like **Amazon, Walmart**, or niche brands with loyal customers.
- ❖ **Cybersecurity Risks:** Threats like data breaches or online fraud can damage customer trust.
- ❖ **Changing Regulations:** Tax laws, data privacy (e.g., GDPR), or shipping restrictions can complicate operations.
- ❖ **Customer Behavior Shift:** Consumers may prefer social commerce (shopping via platforms like Instagram or TikTok), affecting traditional e-commerce traffic.



# Imagine Your E-commerce Presence (cont.)

- ▶ Know yourself—SWOT analysis
- ▶ Develop an e-commerce presence map  
(Fig 4-2)
- ▶ Develop a timeline: Milestones
- ▶ How much will this cost?
  - ▶ Simple Web sites: up to \$5000
  - ▶ Small Web start-up: \$25,000 to \$50,000
  - ▶ Large corporate site: \$100,000+ to millions

# E-commerce Presence Map



# Building an E-commerce Site: A Systematic Approach



- ▶ Most important management challenges:
  - ▶ Developing a clear understanding of business objectives
  - ▶ Knowing how to choose the right technology to achieve those objectives



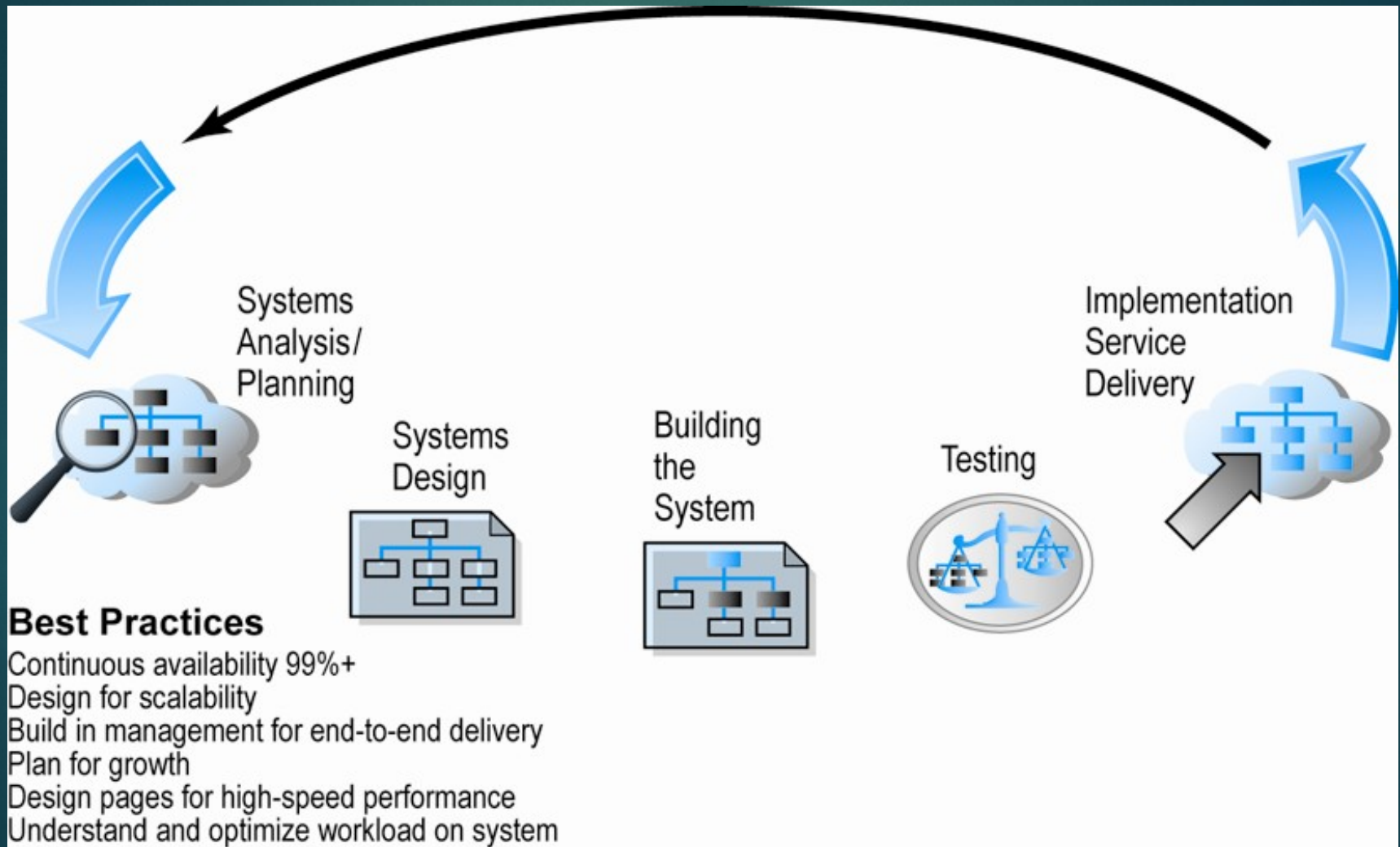
# Pieces of the Site-Building Puzzle

- ▶ Main areas where you will need to make decisions:
  - ▶ Human resources and organizational capabilities
    - ▶ Creating team with skill set needed to build and manage a successful site
  - ▶ Hardware/Software
  - ▶ Telecommunications
  - ▶ Site design

# The Systems Development Life Cycle

- ▶ Methodology for understanding business objectives of a system and designing an appropriate solution
- ▶ Five major steps:
  - ▶ Systems analysis/planning
  - ▶ Systems design
  - ▶ Building the system
  - ▶ Testing
  - ▶ Implementation

# Web Site Systems Development Life Cycle





# System Analysis/Planning

- ▶ Business objectives:
  - ▶ List of capabilities you want your site to have
- ▶ System functionalities:
  - ▶ List of information system capabilities needed to achieve business objectives
- ▶ Information requirements:
  - ▶ Information elements the system must produce in order to achieve business objectives

**TABLE 4.2****SYSTEM ANALYSIS: BUSINESS OBJECTIVES, SYSTEM FUNCTIONALITIES, AND INFORMATION REQUIREMENTS FOR A TYPICAL E-COMMERCE SITE**

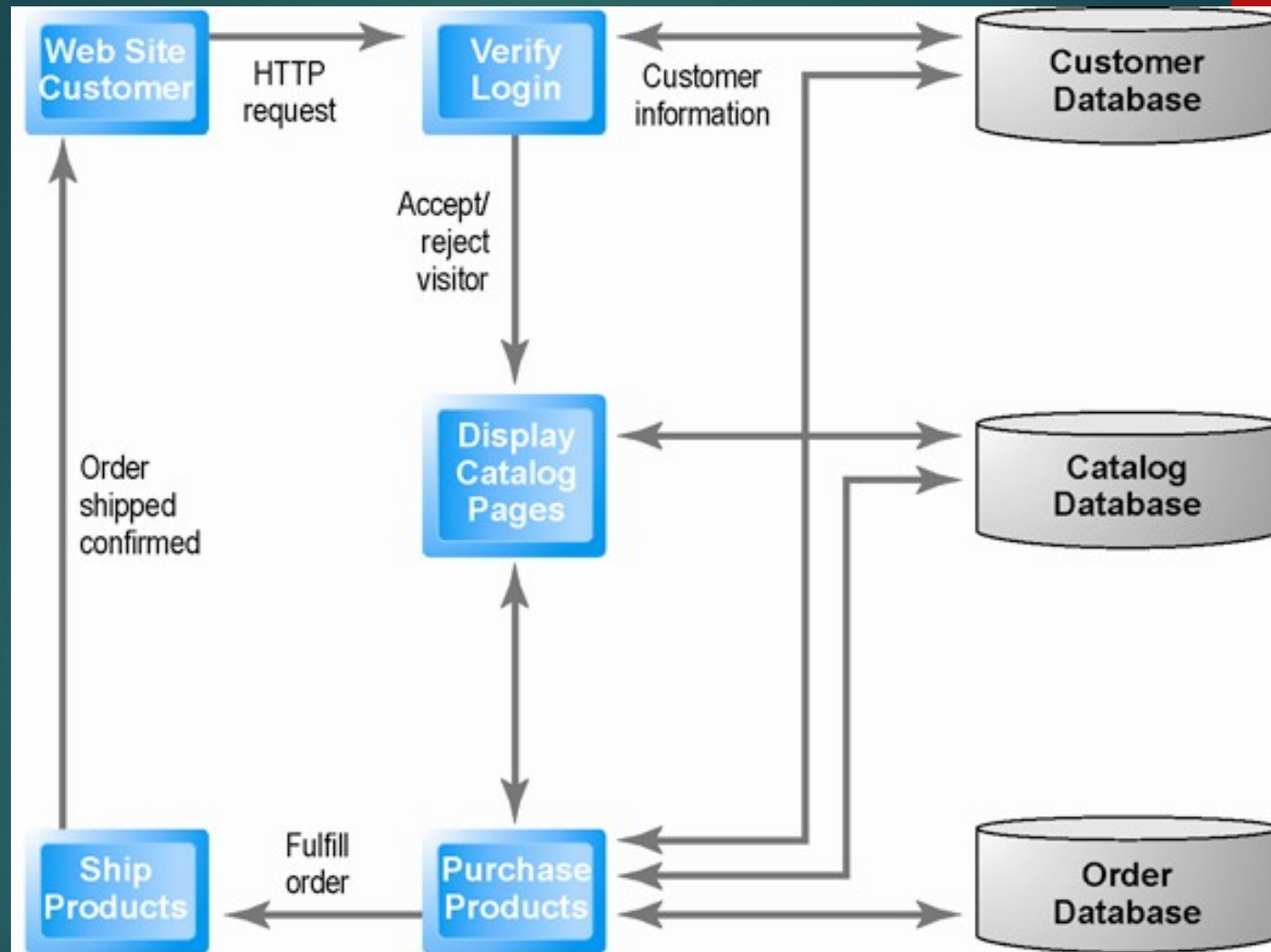
| BUSINESS OBJECTIVE                    | SYSTEM FUNCTIONALITY                                                  | INFORMATION REQUIREMENTS                                                                                              |
|---------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Display goods                         | Digital catalog                                                       | Dynamic text and graphics catalog                                                                                     |
| Provide product information (content) | Product database                                                      | Product description, stocking numbers, inventory levels                                                               |
| Personalize/customize product         | Customer on-site tracking                                             | Site log for every customer visit; data mining capability to identify common customer paths and appropriate responses |
| Engage customers in conversations     | On-site blog                                                          | Software with blogging and community response functionality                                                           |
| Execute a transaction                 | Shopping cart/payment system                                          | Secure credit card clearing; multiple payment options                                                                 |
| Accumulate customer information       | Customer database                                                     | Name, address, phone, and e-mail for all customers; online customer registration                                      |
| Provide after-sale customer support   | Sales database                                                        | Customer ID, product, date, payment, shipment date                                                                    |
| Coordinate marketing/advertising      | Ad server, e-mail server, e-mail, campaign manager, ad banner manager | Site behavior log of prospects and customers linked to e-mail and banner ad campaigns                                 |
| Understand marketing effectiveness    | Site tracking and reporting system                                    | Number of unique visitors, pages visited, products purchased, identified by marketing campaign                        |
| Provide production and supplier links | Inventory management system                                           | Product and inventory levels, supplier ID and contact, order quantity data by product                                 |

# Systems Design: Hardware and Software Platforms

- ▶ System design specification:
  - ▶ Description of main components of a system and their relationship to one another
- ▶ Two components of system design:
  - ▶ Logical design
    - ▶ Data flow diagrams, processing functions, databases
  - ▶ Physical design
    - ▶ Specifies actual physical, software components, models, and so on



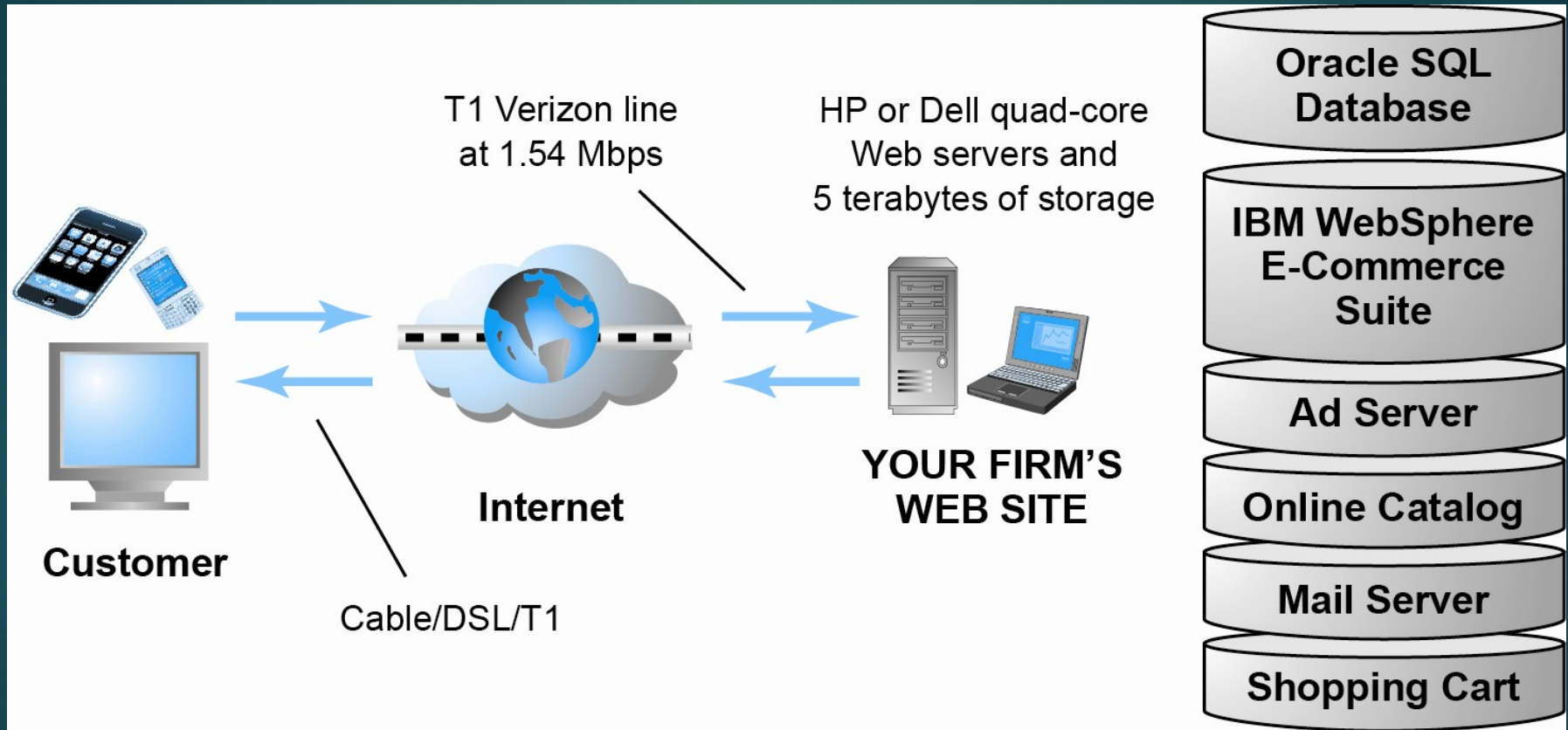
# Logical Design for a Simple Web Site



**(a) Simple Data Flow Diagram**

This data flow diagram describes the flow of information requests and responses for a sample Web site

# Physical Design for a Simple Web Site

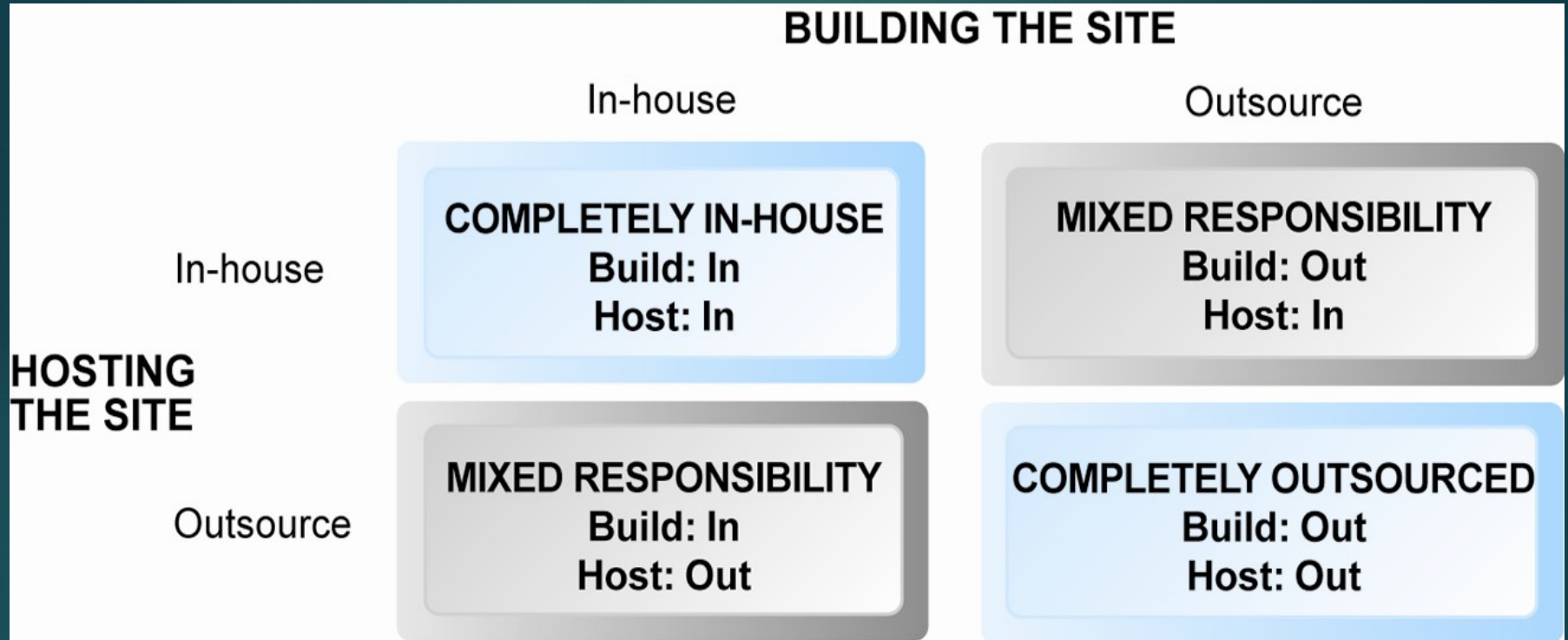


# Build/Host Your Own vs. Outsourcing

- ▶ Outsourcing: Hiring vendors to provide services involved in building site
- ▶ Build own vs. outsourcing:
  - ▶ Build your own requires team with diverse skill set; choice of software tools; both risks and possible benefits
- ▶ Host own vs. outsourcing
  - ▶ **Hosting:** Hosting company responsible for ensuring site is accessible 24/7, for monthly fee
  - ▶ **Co-location:** Firm purchases or leases Web server (with control over its operation), but server is located at vendor's facility



# Choices in Building and Hosting



# Testing, Implementation, and Maintenance

- ▶ Testing
  - ▶ Unit testing
  - ▶ System testing
  - ▶ Acceptance testing
- ▶ Implementation and maintenance:
  - ▶ Maintenance is ongoing
  - ▶ Maintenance costs: Similar to development costs
  - ▶ Benchmarking

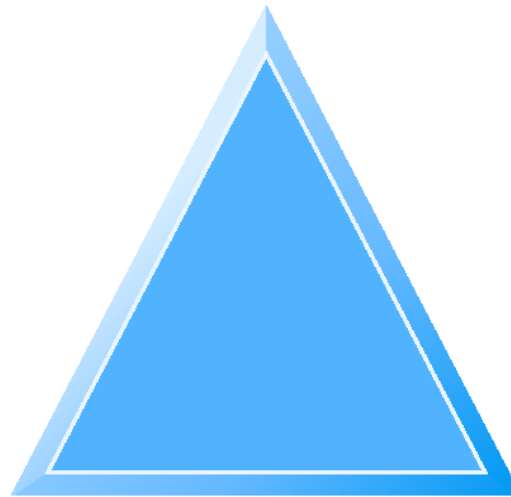
# Factors in Web Site Optimization

## Page Delivery

Content delivery networks  
Edge caching  
Bandwidth

## Page Generation

Server response time  
Device-based accelerators  
Efficient resource allocation  
Resource utilization thresholds  
Monitoring site performance



## Page Content

Optimize HTML  
Optimize images  
Site architecture  
Efficient page style



# Simple vs. Multi-tiered Web Site Architecture

## ▶ System architecture

- ▶ Arrangement of software, hardware, and tasks in an information system needed to achieve a specific functionality

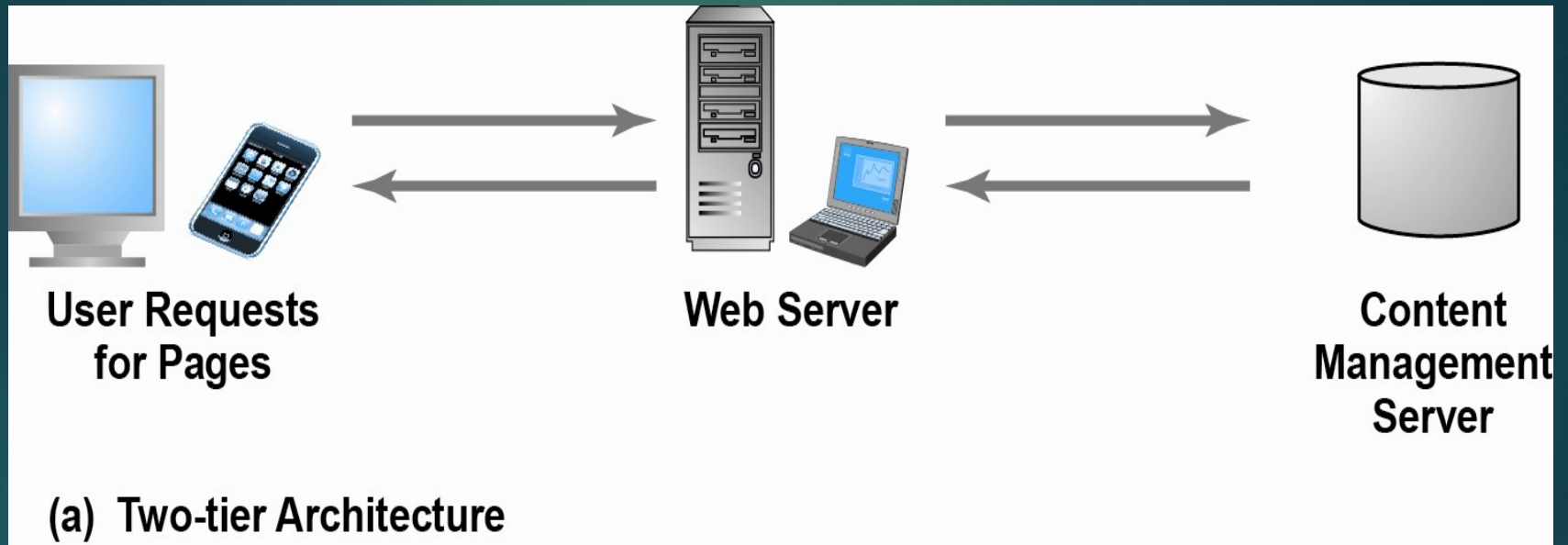
## ▶ Two-tier

- ▶ Web server and database server

## ▶ Multi-tier

- ▶ Web application and other servers
- ▶ Backend, legacy databases

# Two-Tier E-commerce Architecture



# Multi-Tier E-commerce Architecture

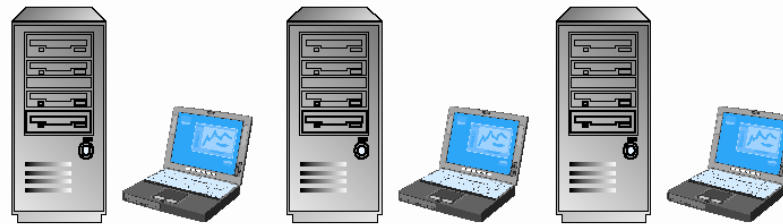
## Web Server Layer

Incoming Internet requests  
T1 Line  
1.544 Mbps



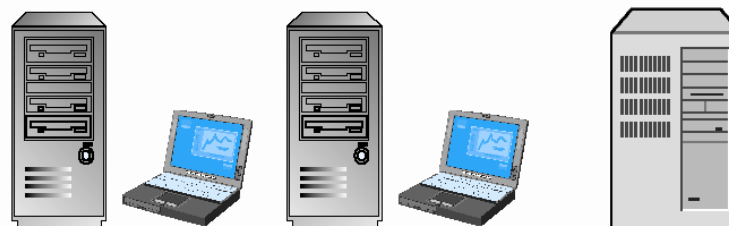
Web Servers

## Middle-tier Layer



E-commerce Servers  
Application Servers  
Database Servers  
Ad Servers  
Mail Servers

## Backend Layer



Corporate applications  
Finance  
Production MRP  
Enterprise systems  
HR systems

## (b) Multi-tier Architecture

In a multi-tier architecture, a Web server is linked to a middle-tier layer that typically includes a series of application servers that perform specific tasks, as well as to a backend layer of existing corporate systems.

# Web Server Software

## ▶ Apache

- ▶ Leading Web server software (47% of market)
- ▶ Works with UNIX, Linux operating systems
- ▶ Comes loaded on IBM systems

## ▶ Microsoft's Internet Information Server (IIS)

- ▶ Second major Web server software (20% of market)
- ▶ Windows-based



**TABLE 4.4****BASIC FUNCTIONALITY PROVIDED BY WEB SERVERS**

| FUNCTIONALITY                                                         | DESCRIPTION                                                                                                                                            |
|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Processing of HTTP requests                                           | Receive and respond to client requests for HTML pages                                                                                                  |
| Security services (Secure Sockets Layer)/<br>Transport Layer Security | Verify username and password; process certificates and private/public key information required for credit card processing and other secure information |
| File Transfer Protocol                                                | Permits transfer of very large files from server to server                                                                                             |
| Search engine                                                         | Indexing of site content; keyword search capability                                                                                                    |
| Data capture                                                          | Log file of all visits, time, duration, and referral source                                                                                            |
| E-mail                                                                | Ability to send, receive, and store e-mail messages                                                                                                    |
| Site management tools                                                 | Calculate and display key site statistics, such as unique visitors, page requests, and origin of requests; check links on pages                        |

# Site Management Tools

- ▶ Basic tools
  - ▶ Included in all Web servers
    - ▶ Verify that links on pages are still valid
    - ▶ Identify orphan files
- ▶ Third-party software for advanced management
  - ▶ Monitor customer purchases, marketing campaign effectiveness, and so on
  - ▶ WebTrends Analytics 10, Google Analytics

# Dynamic Page Generation Tools

## ► Dynamic page generation:

- Page contents stored in databases as objects and fetched when needed
- Lowers menu cost (cost of changing the price of goods/services)
- Enables market segmentation

## ► Common tools:

- Common Gateway Interface (CGI) (offers a standard protocol for web servers to execute programs)
- Active server pages (ASP) (Microsoft's server-side script engine that enables dynamically-generated web pages)
- Java Server Pages (JSP) (same purpose as above)
- Open Data Base Connectivity (ODBC), a std DB access method allows connections to any DB

## ► Advantages

- Lowers menu costs
- Permits easy online market segmentation
- Enables cost-free price discrimination
- Enables content management system (CMS)

# Application Servers

- ▶ Web application servers:
  - ▶ Is a server program in a distributed network that provides the business logic or functionality required by a website. Often viewed as a 3 tier application.
  - ▶ It divides the application into
    - ▶ 1<sup>ST</sup> tier: front end – web browser GUI
    - ▶ 2<sup>nd</sup> middle tier: business logic
    - ▶ 3<sup>rd</sup> tier: back end – DB and transaction server



# E-commerce Merchant Server Software

- ▶ Provides basic functionality for online sales
  - ▶ **Online catalog**
    - ▶ List of products available on Web site
  - ▶ **Online shopping cart**
    - ▶ Allows shoppers to set aside, review, edit selections, and then make purchase
  - ▶ **Credit card processing**
    - ▶ Typically works in conjunction with shopping cart
    - ▶ Verifies card and puts through credit to company's account at checkout

# Merchant Server Software Packages

- ▶ Integrated environment that includes the functionality needed for ecommerce
- ▶ Key factors in selecting a package
  - ▶ Functionality
  - ▶ Support for different business models
  - ▶ Business process modeling tools
  - ▶ Visual site management and reporting
  - ▶ Performance and scalability
  - ▶ Connectivity to existing business systems
  - ▶ Compliance with standards
  - ▶ Global and multicultural capability
  - ▶ Local sales tax and shipping rules

# Web Services and Open-Source Options

- ▶ Options for small firms
  - ▶ Hosted e-commerce sites
    - ▶ Offer site building tools and templates
    - ▶ Example: Yahoo's Merchant Solutions
  - ▶ Open-source merchant server software
    - ▶ Enables you to build truly custom sites
    - ▶ Requires programmer with expertise, time
  - ▶ See table: OPEN SOURCE SOFTWARE OPTIONS

# The Hardware Platform



- ▶ Hardware platform:
  - ▶ Underlying computing equipment needed for e-commerce functionality
- ▶ Objective:
  - ▶ Enough platform capacity to meet peak demand without wasting money
- ▶ Important to understand the factors that affect speed, capacity, and scalability of a site



# Right-Sizing Your Hardware Platform: The Demand Side

- ▶ **Customer demand:**

- ▶ Most important factor affecting speed of site

- ▶ **Factors in overall demand:**

- ▶ Number of simultaneous users in peak periods
  - ▶ Nature of customer requests (user profile)
  - ▶ Type of content (dynamic vs. static Web pages)
  - ▶ Required security
  - ▶ Number of items in inventory
  - ▶ Number of page requests (Google has 40,000 per sec)
  - ▶ Speed of legacy applications
- ▶ SEE TABLE: FACTORS IN RIGHT-SIZING E-COMMERCE PLATFORM

# Right-Sizing Your Hardware Platform: The Supply Side

- ▶ **Scalability:**

- ▶ Ability of site to increase in size as demand warrants

- ▶ Ways to scale hardware:

- ▶ **Vertically**

- ▶ Increase processing power of individual components – e.g., using multiple processors, faster chips

- ▶ **Horizontally**

- ▶ Employ multiple computers to share workload

- ▶ **Improve processing architecture**

**TABLE 4.8****VERTICAL AND HORIZONTAL SCALING TECHNIQUES**

| TECHNIQUE                     | APPLICATION                                                        |
|-------------------------------|--------------------------------------------------------------------|
| Use a faster computer         | Deploy edge servers, presentation servers, data servers, etc.      |
| Create a cluster of computers | Use computers in parallel to balance loads.                        |
| Use appliance servers         | Use special-purpose computers optimized for their task.            |
| Segment workload              | Segment incoming work to specialized computers.                    |
| Batch requests                | Combine related requests for data into groups, process as group.   |
| Manage connections            | Reduce connections between processes and computers to a minimum.   |
| Aggregate user data           | Aggregate user data from legacy applications in single data pools. |
| Cache                         | Store frequently used data in cache rather than on the disk.       |

**TABLE 4.9****IMPROVING THE PROCESSING ARCHITECTURE OF YOUR SITE**

| ARCHITECTURE IMPROVEMENT                        | DESCRIPTION                                                                                       |
|-------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Separate static content from dynamic content    | Use specialized servers for each type of workload.                                                |
| Cache static content                            | Increase RAM to the gigabyte range and store static content in RAM.                               |
| Cache database lookup tables                    | Use cache tables used to look up database records.                                                |
| Consolidate business logic on dedicated servers | Put shopping cart, credit card processing, and other CPU-intensive activity on dedicated servers. |
| Optimize ASP code                               | Examine your code to ensure it is operating efficiently.                                          |
| Optimize the database schema                    | Examine your database search times and take steps to reduce access times.                         |



# Other E-commerce Site Tools

- ▶ Web site design: Basic business considerations
  - ▶ Enabling customers to find and buy what they need
- ▶ Tools for Web site optimization
  - ▶ Search engine placement
    - ▶ Metatags and titles, help search engines find you
    - ▶ Identify market niches, and tailor site to them
    - ▶ Offer expertise such as white papers, industry analysis etc. to attract customers
    - ▶ Encourage links from other sites to yours
    - ▶ Buy Search engine keywords and ads
    - ▶ Use key words to suggest the location of your establishment

**TABLE 4.10****E-COMMERCE WEB SITE FEATURES THAT ANNOY CUSTOMERS**

- |                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"><li>• Requiring user to view ad or Flash introduction before going to Web site content</li><li>• Pop-up and pop-under ads and windows</li><li>• Too many clicks to get to the content</li><li>• Links that don't work</li><li>• Confusing navigation; no search function</li><li>• Requirement to register and log in before viewing content or ordering</li><li>• Slow loading pages</li><li>• Content that is out of date</li></ul> | <ul style="list-style-type: none"><li>• Inability to use browser's Back button</li><li>• No contact information available (Web form only)</li><li>• Unnecessary splash/flash screens, animation, etc.</li><li>• Music or other audio that plays automatically</li><li>• Unprofessional design elements</li><li>• Text not easily legible due to size, color, format</li><li>• Typographical errors</li><li>• No or unclear returns policy</li></ul> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**TABLE 4.11****THE EIGHT MOST IMPORTANT FACTORS IN SUCCESSFUL  
E-COMMERCE SITE DESIGN**

| FACTOR                      | DESCRIPTION                                                                         |
|-----------------------------|-------------------------------------------------------------------------------------|
| Functionality               | Pages that work, load quickly, and point the customer toward your product offerings |
| Informational               | Links that customers can easily find to discover more about you and your products   |
| Ease of use                 | Simple fool-proof navigation                                                        |
| Redundant navigation        | Alternative navigation to the same content                                          |
| Ease of purchase            | One or two clicks to purchase                                                       |
| Multi-browser functionality | Site works with the most popular browsers                                           |
| Simple graphics             | Avoids distracting, obnoxious graphics and sounds that the user cannot control      |
| Legible text                | Avoids backgrounds that distort text or make it illegible                           |

# Tools for Interactivity and Active Content

- ▶ CGI (Common Gateway Interface)
- ▶ ASP (Active Server Pages)/ASP.NET
- ▶ Java, JSP, and JavaScript (Sun)
- ▶ ActiveX and VBScript (Microsoft)
  - ▶ ActiveX are technologies for content downloaded from a network, particularly from the World Wide Web.
- ▶ ColdFusion (Adobe) rapid web application development platform
- ▶ Web 2.0 design elements:
  - ▶ Widgets (small pieces of programming code)
  - ▶ mashups (functionality and data from one program used in another)



# Personalization Tools

- ▶ Personalization
  - ▶ Ability to treat people based on personal qualities and prior history with site
- ▶ Customization
  - ▶ Ability to change the product to better fit the needs of the customer
- ▶ Cookies
  - ▶ Primary method to achieve personalization
    - ▶ Amazon.com customizes the content of the homepage based on products you have viewed in the past

# The Information Policy Set

- ▶ Privacy policy

- ▶ Set of public statements declaring how site will treat customers' personal information that is gathered by site

- ▶ Accessibility rules

- ▶ Set of design objectives that ensure disabled users can affectively access site

# Designing for Accessibility



- ▶ Why might some merchants be reluctant to make their Web sites accessible to disabled Americans?
- ▶ How can Web sites be made more accessible?
- ▶ Should all Web sites be required by law to provide “equivalent alternatives” for visual and sound content?
- ▶ What additional accessibility problems do mobile devices pose?

# Developing a Mobile Web Site and Building Mobile Applications

## ▶ Three types of m-commerce software

- ▶ Mobile Web site
  - ▶ Browser-based Internet services accessed from handheld **mobile** devices
- ▶ Mobile Web app built for a mobile web browser of a smart device (phone, tablet)
- ▶ Native app designed specifically for mobile device

## ▶ Planning and building mobile presence

- ▶ Use systems analysis/design to identify unique and specific business objectives



**TABLE 4.13****UNIQUE FEATURES THAT MUST BE TAKEN INTO ACCOUNT  
WHEN DESIGNING A MOBILE WEB PRESENCE**

| FEATURE      | IMPLICATIONS FOR MOBILE PLATFORM                                                                                                                                                                      |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hardware     | Mobile hardware is smaller, and there are more resource constraints in data storage and processing power.                                                                                             |
| Connectivity | The mobile platform is constrained by slower connection speeds than desktop Web sites.                                                                                                                |
| Displays     | Mobile displays are much smaller and require simplification. Some screens are not good in sunlight.                                                                                                   |
| Interface    | Touch-screen technology introduces new interaction routines different from the traditional mouse and keyboard. The mobile platform is not a good data entry tool but can be a good navigational tool. |

# Developing a Mobile Web Presence

- ▶ Design considerations
  - ▶ Platform constraints: Smartphone/tablet
- ▶ Performance and cost
  - ▶ Mobile Web site:
    - ▶ Least expensive
  - ▶ Mobile app:
    - ▶ Can utilize browser API
  - ▶ Native app:
    - ▶ Most expensive; requires more programming

# Building a Mobile Presence

- ▶ What are the key differences between user experience on a Web site and on a mobile device?
- ▶ Why would a mobile Web site or app from the same merchant need different content or functionality?
- ▶ In which cases would a merchant want to develop a mobile app over a mobile Web site?